MG1CA030 Yamaha 1-cylinder HW kick off meeting

Abbur Thammanna Gowda Pruthviraj (MS/EEF53-PS)

07-02-2023

BOSCH

MG1CA030 Yamaha 1-cylinder HW kick off

Agenda

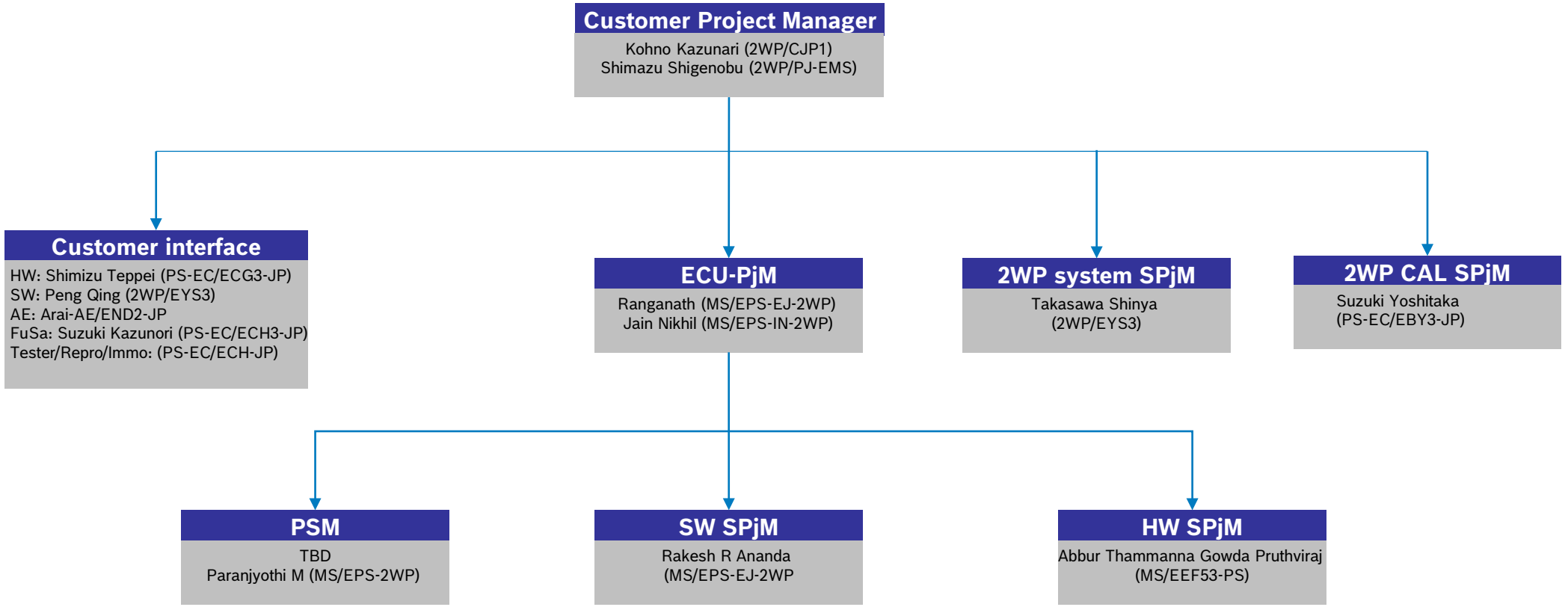
- Org. chart
- General project overview
- Project brief – MG1CA030
- MG1CA030 architecture
- Schedule
- Risk list

Internal | MS/EEF54-PS | 2023-02-06
© Robert Bosch GmbH 2023. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.

BOSCH

MG1CA030 Yamaha 1-cylinder HW kick off

Org chart



Internal | MS/EEF54-PS | 2023-02-06
© Robert Bosch GmbH 2023. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.

MG1CA030 Yamaha 1-cylinder HW kick off

Org chart-HW internal

Sample shop
RBAI, SI

Production plant
RBAI

Project Size Medium

QGC0,QGC2,,QGC4, QGC5 planned

Technical Expert
Santosh Hosangadi (MS/EEF-EP-PS)

HW SPjM
Abbur Thammanna Gowda Pruthviraj (MS/EEF53-PS)

Support PjM
Kuruva Chatanahalli Keshavamurthy (MS/EEF54-MS)

EMI/EMC validation
MS/ENX
Govarthanan D (MS/ENX)

Verification &Validation
MS/EEM-PS
Sheshagiri G K (MS/EEM52-PS)

AE SpJM
Thanikapoikayil Mathew Thomas (MS/EAX17)

Module team
MS/EEH-PS & PS-EC/EHM
Subramanya Vinay (MS/EEH38-PS)

HW Integrator
Harikrishna Surisetty (MS/EEF5-PS)

CAE team
MS/EEH1-PS
Umarani Nandagopal (MS/EEH15-PS)

Layout
MS/EEJ11-AE
Hebbar Preethi Purushothama (MS/EEJ11-AE)

Reliability expert
MS/EAX13
Sughosh P Ramprasad (MS/EAX13)

VALT validation
MS/EEM
Mithun Karadi (MS/EEM51-PS)

© Robert Bosch GmbH 2023. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.

BOSCH

MG1CA030 Yamaha 1-cylinder HW kick off

General project overview

- **ECU Name:** MG1CA030
- **Model Name:** TBD New model
- **Description:** 08W-330cc & 070-200cc 1cylinder engine projects w/ 2 vehicle variants
- **Lead variant :** 08W – CVT Scooter
- **Customer Production:** IYM(IN)/YIMM(ID)
- **Customer Development:** YMC (JP)
- **Customer P/N:** TBD
- **Target Market:** IN and ID
- **Product Requested:** FI-ECU
- **SOP:** 03.26' (CVT) & TBD (MT)
- **Production Location:** NhP (IN)



08W - CVT Scooter
Reference model AEROX



070 - MT Sport
Reference model: R15

Volumes scenario:

Volume	MT車	CVT車1	CVT車2	TOTAL
Del. Loc.	IYM	YIMM	YIMM	
Country	India	Indonesia		
2026年	100,000	31,000		131,000
2027年	100,000	31,000	90,200	221,200
2028年	100,000	31,000	90,200	221,200

Internal | MS/EEF54-PS | 2023-02-06
© Robert Bosch GmbH 2023. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.

MG1CA030 Yamaha 1-cylinder HW kick off

Variant overview– MG1CA030

Criteria	Project 08W	Project 070
Vehicle Type	Scooter	Motorcycle
Transmission	CVT (Mechanical)	MT
No. Cylinder	1	1
Displacement	330cc	200cc
Throttle Type	Electronic Throttle	Electronic Throttle
Emission requirement	EU5 (Details TBD)	EU5 (Details TBD)
OBD requirements	OBD2-2 (2xO2 system)	OBD2-2 (2xO2 system)
Trigger Wheel configuration	12-1 (15-15)	12-1 (15-15)
Features	Cruise control/Var. Speed limiter Ride modes Traction control Head lamp relay control ECO Lamp Immobilizer with YMC specific protocols Hardwired communication with ABS Tester / Immobilizer (K-LINE) EOL requirements (YMC specific) CVT Specific: Generator control, screen motor, seat/grip heater	Cruise control/Var. Speed limiter Ride modes Traction control Head lamp relay control ECO Lamp Immobilizer with YMC specific protocols Hardwired communication with ABS Tester / Immobilizer (K-LINE) EOL requirements (YMC specific) MT Specific: ISG control,
Others requirements	ISO26262 compliant	ISO26262 compliant



08W - CVT Scooter
Reference model AEROX



070 - MT Sport
Reference model: R15

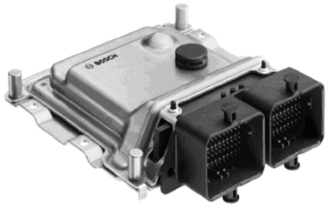
Future requirement
HW support requested

Direct driven Fuel pump (BLDC)
Wide band lambda sensor
Bypass air solenoid
Variable Valve Actuation

Internal | MS/EEF54-PS | 2023-02-06
© Robert Bosch GmbH 2023. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.

MG1CA030 Yamaha 1-cylinder HW kick off

Project brief – MG1CA030



Project details:

- ECU : MG1CA030, Medium project
- Customer : Yamaha Motor compony (YMC)
- Housing: SKA0910
- Pin count: 112pin (2 X 56Pin)
- Series production plant: RBAI
- Proto sample : Si
- Validation and customer sample : RBAI

Top Risks / Problems:

- Technical complexity for ECU architecture in SKA housing.
- Complexity in layout with respect to the requirement.
- Integration/Design of various module for requirement fulfilment.
- Customer C-sample delivery timeline (Nov-2023).

Project Delivery Milestone:

Description	Quantity	Planned
Samples- C samples with prototype PCB	TBD	Nov/2023
Samples- C samples with series PCB	TBD	March/2024

Connected iGPM MCR-IDs:

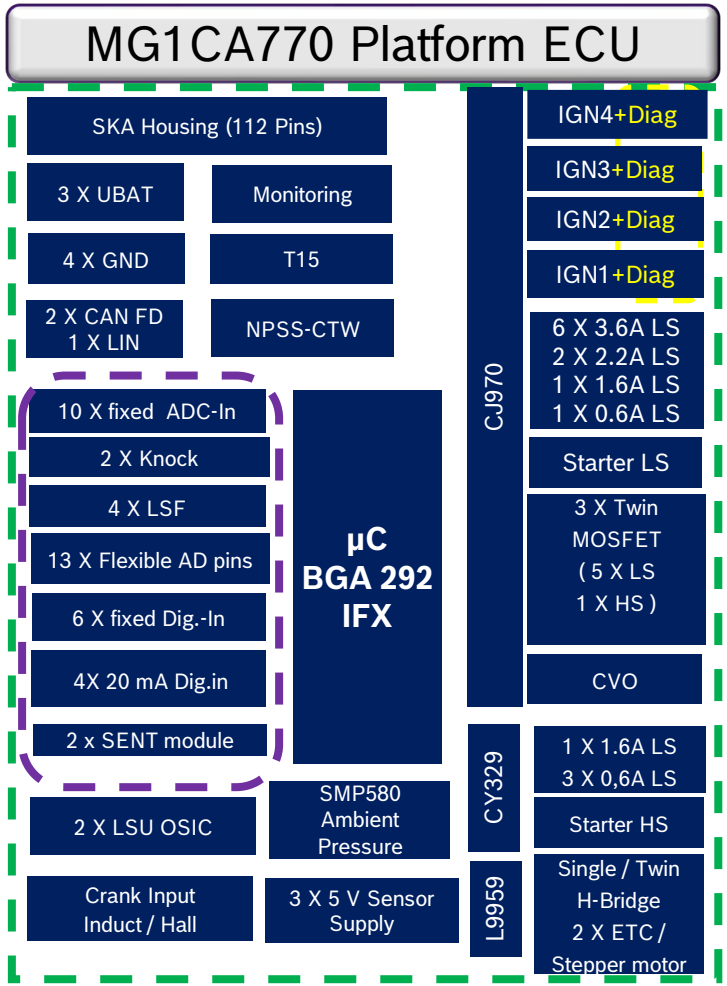
BM-00078422_002 (L2_ECU_HW_YMC_1cyl_MG1; Work) → -

Internal | MS/EEF54-PS | 2023-02-06
© Robert Bosch GmbH 2023. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.

BOSCH

MG1CA030 Yamaha 1-cylinder HW kick off

MG1CA030 Architecture



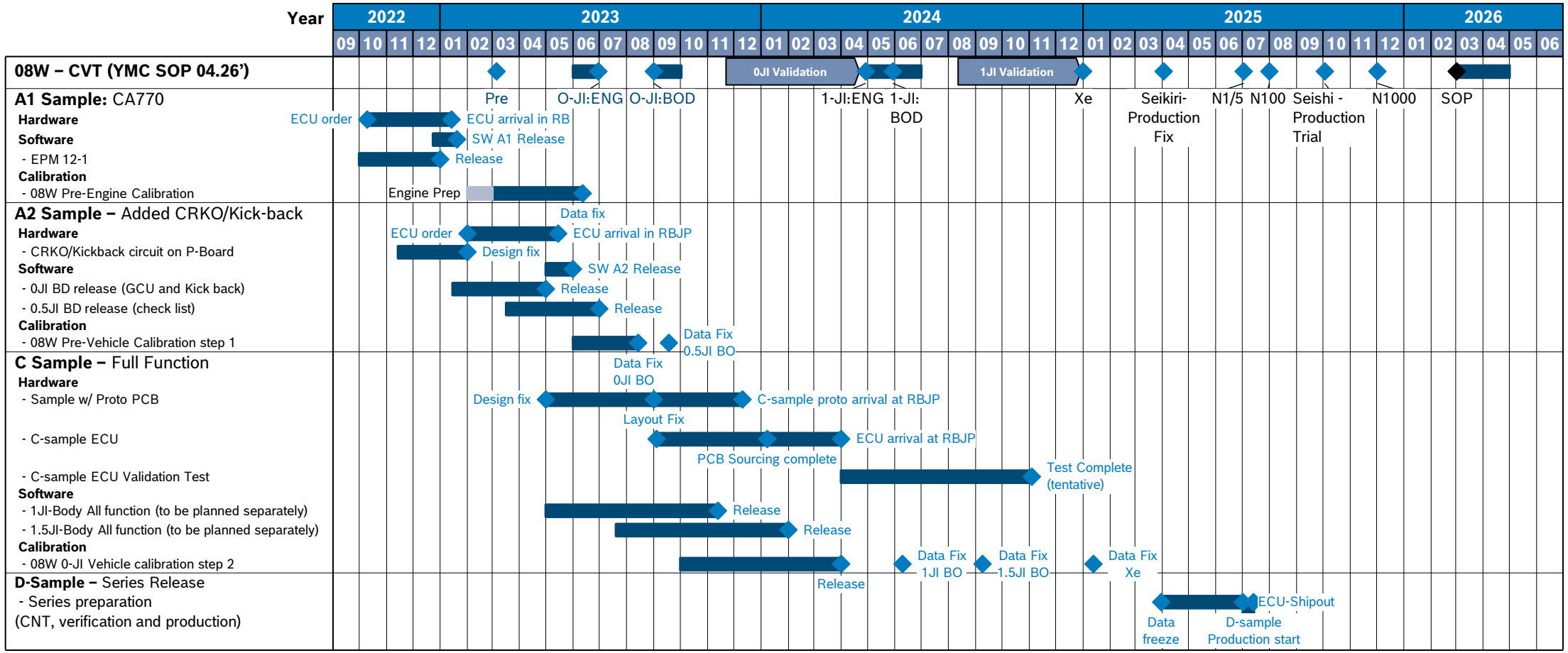


Internal | MS/EEF54-PS | 2023-02-06
© Robert Bosch GmbH 2023. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.

BOSCH

MG1CA030 Yamaha 1-cylinder HW kick off

Series Development Schedule – PRELIMINARY Hardware



Internal | MS/EEF54-PS | 2023-02-06
© Robert Bosch GmbH 2023. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.

BOSCH

MG1CA030 Yamaha 1-cylinder HW kick off

Risk list

Sl.No.	Category	(Sub-) proj.	Risk Source "If"	Risk Impact "Then"	Rating	Measures		Current planned date	Status	Result / Remarks
						Mitigation (Avoidance; Reduction; Transfer)	Contingency Plan			
1	Timeline	Project	❖Requirement freeze	❖Project milestone and SOP date	High	❖Advance planning	❖Requirement freeze ❖Management support		Open	
2	Integration	HW	❖Unreleased and new module in for Ignition slow ON/OFF , CRKO circuit	❖Leads to change in design and PCB layout	High				Open	
3	Layout	HW	❖Present considered requirement results to the increase in component count on limited PCB space.	❖Dense in PCB population. ❖No more additional requirement can be considered. ❖Layout routing complexity increase.	Medium	❖Remove low priority requirement from design			Open	
4	Design	HW	❖Power dissipation	❖Leads to ECU pin failure and PCB trace	Medium	❖Robust design review	❖Power dissipation estimation		Open	
5	Resource	HW	❖PCB resource	❖Project milestone and SOP date	Low	❖Advance planning			Open	
6	Software	SW	❖Availability of the software	❖Leads to delay in sample manufacturing.	Medium					
7	Software	SW	IFX micro controller change over to ST controller	Project milestone and SOP date	High	Advance planning	Management support		Open	Due to software development for ST controller,
8	Software	SW	CY30 is implemented	Negative edge detection implementation	Medium	Advance planning			Open	

Internal | MS/EEF54-PS | 2023-02-06
© Robert Bosch GmbH 2023. All rights reserved, also regarding any disposal, exploitation, reproduction, editing, distribution, as well as in the event of applications for industrial property rights.